Validation and AI Use — NVIDIA (NVDA), Project 1

Decision, version, and as-of boundary

Decision / user	Do not initiate at $225.51; keep on watch — for the buy-side investment committee (no position)
Repository	github.com/Jesstran12/NVDIA-Research, commit `84f78ba` (analysis run); the frozen version is HEAD at push (the frozen commit is also printed in `output/visible_output.json` → `run.commit`)
Run timestamp	printed in `output/visible_output.json` → `run.timestamp`
Source cutoffs	Filings to the Q2 FY2027 10-Q (filed 26 Aug 2026); prices to the 23 Sep 2026 close; peer filings to Broadcom's Q3 FY2026 10-Q (10 Sep 2026). Nothing after 23 Sep 2026
Environment	Python 3.13 (3.12+ required; 3.12.14 also verified 19 Oct 2026), standard library for the engine; pandas 2.3, Streamlit 1.66, pytest 8 for the interface and tests (`requirements.txt`); macOS 15

Data and convention ledger

The complete ledger (33 inputs with locator, type, independent check and risk) is `docs/evidence-ledger.csv`; the base-year normalization (one-time items, basis changes, mechanical checks) is `docs/base-year-normalization.md`. The load-bearing rows:

Material input	Source	As-of	Definition / units	Transformation / exclusion	Risk
Revenue, gross profit, opex, operating income, net income FY2026	10-K FY2026 p. 51 (0001045810-26-000021)	25 Jan 2026	GAAP, USD millions	base year of the pro-forma; H20 charge noted (2.1 pp of gross margin)	provider revenue tag stale since FY2022
Opening balance sheet	10-K FY2026 p. 53	25 Jan 2026	USD millions	reconciled line by line; totals 206,803 / 49,510 / 157,293	none
Bridge: cash 10,605; marketable debt 39,065; marketable equity 12,886; debt 8,468	10-K p. 53; 10-Q comparative column; Note 11	25 Jan 2026	carrying amounts	non-marketable securities (22,251) excluded; leases not debt	marketable equity switchable (–$0.53)
H1 FY2027 actuals; Q3 outlook	10-Q Q2 FY2027 p. 3; 8-K 26 Aug 2026	26 Jul 2026	GAAP; guidance	FY2027E = H1 + Q3 guide + Q4 judgment	Q4 is judgment (±10% bound)
Shares 24,100m + 95m dilution	10-Q cover (21 Aug 2026); Note 4	Aug 2026	basic outstanding + dilutive increment	diluted 24,195m	cover count rounded to 0.1bn
Price $225.51	Nasdaq close cached in `src/beta_prices/`	23 Sep 2026	unadjusted close	reverse-DCF target; market cap	none
Cost of equity 12%	judgment; `beta.py` 2.21 raw / 1.81 Blume; rf 5.00%, ERP 4.42% (course)	Sep 2026	CAPM cross-check 13.0–14.8%	bounds 9–16%	12% is the generous end
Cost of debt 2.92%	10-Q Note 9 effective rates, principal-weighted over the Jan 2026 series	25 Jan 2026	pre-tax	after-tax at 17%	immaterial (debt weight 0.16%)
Peers: AMD, AVGO trailing GAAP EPS / EBIT, cash, debt, shares	SEC XBRL companyfacts, accessions in the ledger	Jun–Aug 2026	TTM GAAP	policy frozen before the numbers; >100x not meaningful	acquired-intangible amortisation depresses GAAP

Validation register

Claim / output	Failure mode	Test and expected result	Actual result	Disposition	Evidence
The DCF arithmetic is right	discounting, terminal, bridge errors	**Known answer (training case):** FCFF 100; 8/6/5/4/3; 10%; 3%; 50/300/50 → twelve lines incl. PV explicit 448.4408, TV 1,894.6486, EV 1,624.8685, $27.4974, 72.40%	all twelve match to four decimals; WACC 11% → $23.41	pass	`tests/test_project1.py::test_training_case_dcf_known_answer`
The three-statement engine is right	unlinked lines	**Known answer (ABG, Lab 09):** $291.75, FY2026E FCFE 211.4, FY2030E cash 719.8, ~80% terminal	$291.75; 211.4; 719.8; 79.8%	pass	`test_abg_engine_known_answer`; `abg-proforma.md`
The Project 1 wrapper does not change the Lab 10 model	wrapper drift	FCFE through `engine.run_proforma` = $123.05	$123.05	pass	`test_nvda_fcfe_base_reproduced_through_wrapper`
Statements articulate	balance / cash / roll-	every year, every run: assets – liabilities –	all residuals 0.0	pass	`test_statements_articulate_every_year_under_shocks`; app checks panel

Claim / output	Failure mode	Test and expected result	Actual result	Disposition	Evidence
	forward breaks	equity = 0; cash BS = CF; PP&E and debt rolls = 0, under shocks too			
Base year reconciles to the filing	typed opening sheet	model opening totals = 10-K p. 53	gaps 0.0	pass	`test_opening_balance_sheet_reconciles_to_filing`
Independent source / definition check on a load-bearing input	wrong definition	FY2026 revenue 215,938 and FY2026 debt 8,468 read from the filing text, then from the SEC XBRL API (`Revenues`, `LongTermDebt`); marketable securities 51,951 (10-K) decomposed as 39,065 debt + 12,886 equity from the 10-Q comparative column	all agree; decomposition sums exactly	pass	`docs/evidence-ledger.csv`; `src/sec_xbrl/companyfacts_extract.csv`
Directional changed-input test, written before execution	monotonicity	**Prediction:** raising the cost of equity from 12% to 13% lowers value per share by roughly a tenth, because every discount factor shrinks and the terminal denominator widens from 9% to 10%	$123.93 → $111.46 (−10.1%); value strictly falls across 9–16% and rises with terminal growth 0–4%	pass, as predicted	`test_value_falls_as_cost_of_equity_rises`; `output/visible_output.md` grid
Boundary: g ≥ WACC refused	infinite value	`dcf()` returns None at g = WACC and g > WACC; the app shows the refusal in red	None; red check "Terminal growth below the discount rate"	pass	`test_terminal_growth_at_or_above_wacc_is_refused`
Enterprise-to-equity bridge	double counting; wrong shares	EV + cash & securities − debt = equity to the cent; diluted point-in-time shares; bridge dated at the cash-flow start	2,944,373 + 62,556 − 8,468 = 2,998,461; 24,195m; dated 25 Jan 2026	pass	`test_bridge_reconciles_and_uses_diluted_point_in_time_shares`
Peer-policy and multiple-definition consistency	mixed dates / definitions	same price date for all; EV = market cap + carrying debt − cash & STI; TTM GAAP; >100x flagged	consistent; AMD 158x / 153x flagged not meaningful	pass	`test_comps_policy_consistency`
Highest-risk failure test: the FY2028–31 growth path is the valuation	the conclusion rests on one judgment	If the price can be reached inside the defended growth bound, the "do not initiate" call is only as strong as the bound. Reverse DCF: solve the growth shift for $225.51	+23.7 pp a year (path ≈ 49 / 36 / 31 / 28%) — inside the +25 pp bound by 1.3 pp; the high case (+10 pp) gives $187	**qualified:** the call survives, but only because the bound stops one point above the implied path. Reported as the main risk; the trigger table makes it reviewable each quarter	`output/visible_output.md` reverse DCF; `docs/decision-memo.md` §5
Loud failure on a typed cash line	silent error	`python analysis.py --break`: cash typed at the opening value; the model must refuse and name the gap	"Model refused: FY2027E does not balance: assets − liabilities −	pass	`test_broken_cash_line_is_refused`

Claim / output	Failure mode	Test and expected result	Actual result	Disposition	Evidence
			equity = −10,088.0"		
Base restored after every shocked run	state leakage	module assumptions identical before and after sweeps; base value identical	identical; $123.93 both times	pass	`test_base_restored_after_shocked_runs`; `nvda_sensitivity.py` restored-base PASS (difference 0.000000)
Out-of-bounds input is clamped and reported	silent extrapolation	cost of equity 50% → clamped to 16% and listed	clamped, reported	pass	`test_out_of_bounds_input_is_clamped_and_reported`
Ranking depends on the ranges?	false precision in the driver ranking	rank at full bounds, half bounds, Lab 11 ranges	growth path first in all three; middle order changes with the bounds	reported as "over these ranges"	`output/visible_output.md` sensitivity
Cold run (fresh environment, README only)	hidden dependency	clone → venv → `pip install -r requirements.txt` → `pytest` → `python analysis.py` → `streamlit run app.py`	no failure; 17 tests passed; visible output written 22 s after the clone; app answered HTTP 200 (commit `d88cb9a`)	pass	`docs/cold-run-log.md`

Locked Changed-Input Record — human prediction first; AI closed during execution

Precommit before the run and before AI assistance on the prediction

Precommit timestamp	Repository commit	Material input	Old → new value / units	Expected output direction	Expected decision effect
2026-09-29 17:56:29 −04:00	`3bcbdde` ("Lab 11": `nvda-sensitivity.md` with the prediction and an empty Results section)	Revenue growth FY2028E–FY2031E (AI Data Center demand)	25 / 12 / 7 / 4 % → 20 / 7 / 2 / −1 % (−5 pp each year) and 30 / 17 / 12 / 9 % (+5 pp)	value per share falls about 15–20% on the lower path; roughly symmetric on the upper path; growth moves value about three times as much as gross margin	no change to watch/defer; confirms that the growth path after FY2027 is the assumption to research
same	same	Gross margin FY2027E–FY2031E (CUDA pricing power)	74.5 → 70.0 % path → −3 pp and +3 pp each year	value falls about 5% on the lower path (a 3 pp cut on a ~70% margin is ~4% of gross profit; opex does not scale down)	no change

The prediction text is the Locked Changed-Input Record section of `nvda-sensitivity.md` as committed in `3bcbdde`; it was written with AI closed and before `nvda_sensitivity.py` was run.

Preserved execution result

Before / after output	Actual decision effect	Prediction reconciliation	Failure diagnosis or why the action did not change	Evidence path
Growth −5 pp: $123.05 → **$107.55** (−12.6%); FY2031E operating income 359,021 → 298,796; FCFE 307,656 → 260,739. Growth +5 pp: → **$140.44** (+14.1%). Span $32.89	none: watch/defer kept; growth path confirmed as the first research priority and the first slider in the product	Direction right. Order right (growth > margin). Size: the growth effect was smaller than the 15–20% predicted (−12.6%) and the ratio was 2.6× rather than 3×. Symmetry roughly held (−15.50 vs +17.39)	The prediction assumed value falls as much as FY2031 revenue (−17%). It fell less because (1) FY2027E, held fixed, carries about a third of the explicit-period value, and (2) lower growth shrinks the working-capital build, which partly offsets lower profit in FCFE (the FY2028E build falls from 14,075 to 11,483 and the FY2031E build from 4,354 to 266). The margin prediction (−5%) matched −5.1%	`output/lab11_sensitivity_run.txt`; `nvda-sensitivity.md` Results; commit history of `nvda-sensitivity.md`
Gross margin −3 pp: $123.05 → **$116.73** (−5.1%); +3 pp → **$129.37** (+5.1%). Span $12.65	none	matched the prediction	not applicable	same

The Project 1 FCFF engine reproduces the same test (±5 pp growth → $108.48 / $141.25; ±3 pp margin → $117.64 / $130.21), so the record holds under both cash-flow definitions.

Named AI-use register

Dispositions are Jess Tran's. The rows dated 2026-10-19 were drafted by the AI with the evidence it used and adopted by the student after reading the checks; the student owns each verdict and its reason.

Tool / model surface and version	Interaction date	Material task	Output / claim used or considered	Independent check	Accept / modify / qualify / reject	Effect on decision
ChatGPT / Codex (Lab 03 session)	2026-09-03	Draft the Edition A report structure from the student's notes	report wording; UNSOURCED markers	sources gathered and checked by the student in the 10-K	accept (Edition A, before the project boundary)	none
Claude Code (Lab 09–12)	2026-09-24 to 2026-10-01	Generate the ABG engine from the Lab 09 request; port to NVIDIA; sensitivity and what-if scripts	`proforma.py`, `nvda_proforma.py`, `nvda_sensitivity.py`, `archive/nvda_whatif.py`	ABG known answer $291.75; check block 0.0 every year; `--break` refuses	accept after known-answer check	base case $123.05
Claude Code, Claude Fable 5.1	2026-10-19	Build the Project 1 layer (`engine.py`, `analysis.py`, `app.py`, tests, docs)	FCFF DCF, bridge, checks, sensitivities, comps, reverse DCF, the recommendation rule	training-case known answer to four decimals; FCFE cross-check $123.05 vs FCFF $123.93; 17 tests; articulation on every run	accept after the checks	the committee action and range
Claude Code, Claude Fable 5.1	2026-10-19	Bridge balance sheet: AI first proposed the latest 10-Q (26 Jul 2026: cash 22,443; marketable 76,926; debt 33,366)	would have added the $24.9bn June 2026 notes and the H1 FY2027 cash build to the bridge	the pro-forma starts at 25 Jan 2026, so H1 FY2027 cash flow is already inside FY2027E FCFF: the July sheet double counts it (about +$0.99 per share)	**corrected:** bridge at the opening date; the July sheet kept as a flagged reference toggle that turns the checks panel red	none on the action; avoids a false +1%
Claude Code, Claude Fable 5.1	2026-10-19	AI proposed including non-marketable securities (22,251) in the bridge as a non-operating asset	+$0.92 per share	no observable price; strategic stakes; 10-Q Note 7 carries them at cost less impairment	**rejected**	none
Claude Code (Lab 10–12)	2026-09-24 to 2026-10-01	Fresh-eyes challenge reviews in place of a human partner (none was available): "why decelerate when H1 FY2027 ran at +106%?"; "the margin shock understates CUDA"; "what happens to NVIDIA if AI stalls?" (Apple comparison)	the downside case and the growth bounds	each challenge rerun through the engine with the checks passing (low case $80.31)	growth-path challenge rejected as a change; CUDA and AI-stall challenges **accepted**: the low case moves growth, margin, opex and the discount rate together, and –25 pp is the "AI capex stops" bound	sets the low end of the $80–$187 range
Claude Code, Claude Fable 5.1	2026-10-19	AI computed AMD's trailing P/E (158x) and EV/EBIT (153x) and proposed excluding AMD	implied NVIDIA values of $617 / $582	AMD's 15.7% operating margin vs NVIDIA's 65%: the multiple capitalises a recovery, not earnings	**qualified:** AMD stays in the table under the frozen policy, its multiples reported but marked not meaningful under the pre-declared >100x rule	none on the action; the comps disagreement is explained, not averaged
Claude Code, Claude Fable 5.1	2026-10-19	AI drafted the committee action from the rule "price above the high case → do not initiate"	"do not initiate at this price; keep on watch"	the student's standing call since Lab 03 was watch/defer; the high case ($187) is a full rerun with checks passing; the reverse DCF puts the gap at +24 pp growth	accept: the committee hears "do not initiate at this price; keep on watch"	the headline
Claude Code (archive/nvda_whatif.py)	2026-09-24	AI proposed the 10% tutorial cost of equity as a comparison row	$159 per share	measured beta 2.21 → CAPM 13–14.8%	rejected as base; kept as a grid row	12% stays, labelled generous

Private prompts, credentials and trivial completions are not recorded. No credentials exist in this project.

Cold-run log

See `docs/cold-run-log.md` (fresh clone, fresh virtual environment, README instructions only, first failure and time to visible output recorded).

Limitations, monitoring, and kill / escalation rules

- **Kill the base case** if Data Center revenue falls quarter on quarter in either of the next two reports; rebuild as the low case before any committee discussion.
- **Escalate** if the bridge fails to reconcile on the next 10-Q, if the balance sheet check fails on any input inside the bounds, or if the reverse-DCF growth shift moves inside the high case (+10 pp): the "do not initiate" call would then rest on a judgment the evidence no longer supports.
- **Known limitations:** valuation struck at 25 Jan 2026 against a 23 Sep 2026 price (roll-forward about +8% before distributions); 53-week FY2027; SBC add-back (−$6 if treated as cash); terminal reinvestment of 2.5% of net income at 3% growth; the course rf / ERP convention not re-sourced to September 2026; peer GAAP earnings depressed by amortisation; the cover share count rounded to 0.1bn.